COST:

A cost is an expenditure incurred by a firm to produce goods and services for sale in the market. In other words, a cost is the outflow of money from the business to gain inflow of money after sale of the commodity. A producer has to incur various costs in order to produce goods and services. These costs are of various types:

Direct cost or explicit cost:

Explicit costs are those costs which are met by cash payments for employing various factors of production. The producer actually pays money to produce his goods and services. A direct or explicit cost is the material, labour, expenses, overheads, selling and distribution, administrative cost related to production of a commodity.

Indirect cost or implicit cost:

Implicit costs are those costs which the firm lets go or sacrifices in order to hire an alternative factor of production. These costs are opportunity costs of the factors of production. Implicit cost is also called as imputed cost. Here cash outflow does not happen.

SHORT- RUN COSTS OF PRODUCTION:

These broadly comprise of the total, average and marginal costs.

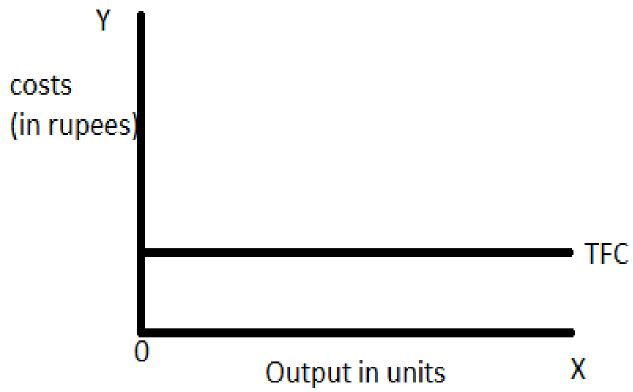
Q	FC	VC	MC	TC	ATC	AVC
1	200	150	150	350	350	150
2	200	250	100	450	225	125
3	200	300	50	500	167	100
4	200	340	40	540	135	85
5	200	400	60	600	120	80
6	200	460	60	660	110	77
7	200	560	100	760	109	80
8	200	700	140	900	113	88
9	200	840	140	1040	116	93
10	200	1100	260	1300	130	110
11	200	1500	400	1700	155	136
12	200	2200	700	2400	200	183

TOTAL COSTS (TC)

It is the sum of total fixed cost and total variable cost **TC =TFC+TVC** and also TC is the actual cost incurred to produce a given quantity of output.

Total Fixed Costs:

refer to the sum of all the expenditures by the firm on fixed inputs like land, machinery, insurance etc. The payments for such factors are fixed in the short run and independent of the level of output. **Even at zero level of production, the firm has to incur fixed costs which remain unchanged at all levels of output.** The TFC curve runs parallel to the X-axis. For e.g. the cost incurred by a firm on fixed machinery, building blocks, remain fixed over a given span of time.

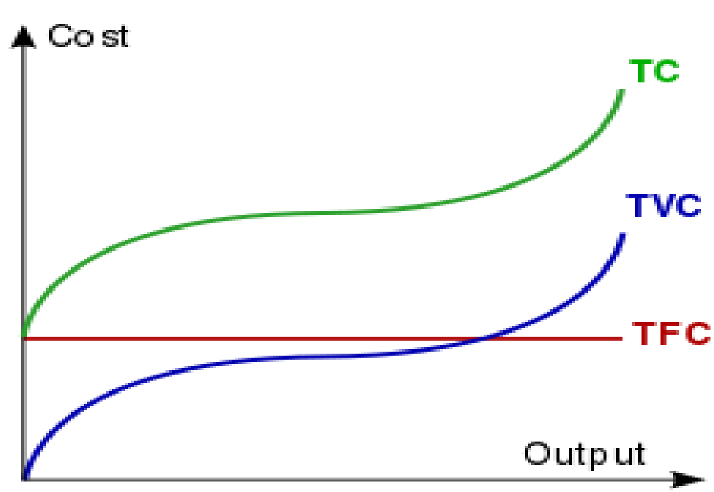


FIXED COST AND SUNK COSTS:

Sunk costs are costs that have been incurred but cannot be retrieved if the firm goes out of business. Fixed costs can be escaped by going out of business. An example of Sunk costs is the expenditure incurred in purchasing a machine which does not have an alternative use when the firm decides to go out of business.

TOTAL VARIABLE COSTS:

Refers to the firm's total expenditure on variable factors. Variable costs change directly with the change in the level of output. Examples of such costs are costs of labor, raw materials, transportation etc. TVC is zero when output is zero. **TVC has an inverse 'S '– shape reflecting the law of variable proportions**



SHORT RUN AVERAGE COSTS (Analysis of Short Run AC curves)

In order to find out the per unit profit, the firm has to make a comparison between the **per unit cost** or the average costs and the per unit price or simply the price. **The Average Cost is the sum of the Average fixed costs (AFC) and the Average variable cost (AVC).**

Average fixed costs (AFC):

AFC is the per unit cost of the fixed factors of production. AFC= TFC/Q. The AFC is a rectangular hyperbola because multiplication of AFC with the quantity of output produced always yields a fixed value. The AFC will never touch the x-axis as the AFC cannot be zero, however large the level of output.

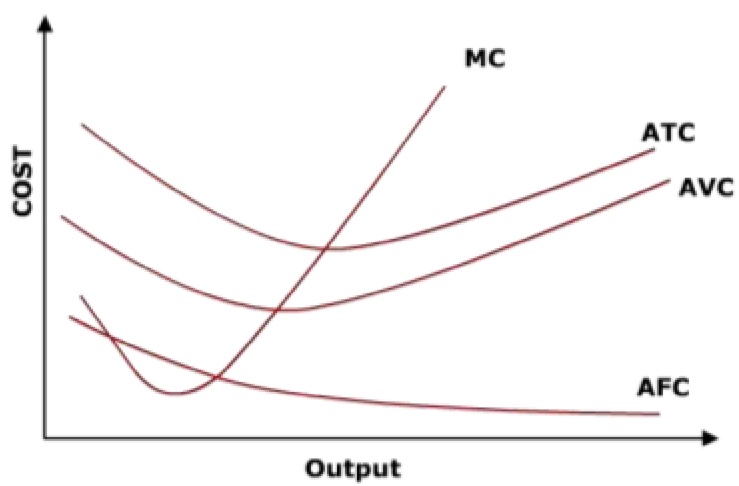
Average variable cost (AVC):

AVC refers to the per unit cost of the variable factors of production. AVC= TVC/Q, where Q is the level of output. Since the total variable costs (TVC) are determined by the law of variable proportions, the AVC falls initially and rises later. The AVC is a dish –shaped curve

RELATIONSHIP BETWEEN AC, AVC and AFC (Analysis of Short Run AC curves):

The U-shape of AVC and AC curves is due to the law of variable proportions. The behavior of the AC curve depends on the behavior of the AVC and AFC curves**. Initially both AFC and AVC are falling leading to the fall in AC.** The minimum point of AC occurs to the right of the minimum point of AVC. After reaching its minimum point, it starts rising. However, the AFC continues to fall. The AC reaches its minimum point when the rate of fall of AFC is equal to the rate of rise of AVC. **When the rate of rise in AVC becomes greater than the rate of fall in AFC, the AC starts rising.**

The vertical distance between AC and AVC is the AFC, which continues to decline as the output increases.



MARGINAL COST (MC)

refers to the incremental cost and is the addition to the total cost as a result of a unit increase in the output.

$MC = \Delta TC/\Delta Q$

Since the fixed cost remains constant in the short-run, the marginal cost is also defined as the increase in total variable cost due to a unit increase in output.

Mathematically,

$MC_n = TC_n – TC_{n-1}$

MC_n = marginal cost of producing the n^{th} unit,

TC_n= total costs of producing the n units

TC_{n-1}= total costs of producing 'n-1 'units

Graphically, the marginal cost is the slope of the total cost curve. With an inverse S-shaped of the total cost curve, the **MC curve is U- shaped. In the short-run, the AC, AVC and MC curves are U-shaped and AFC is a rectangular hyperbola.**

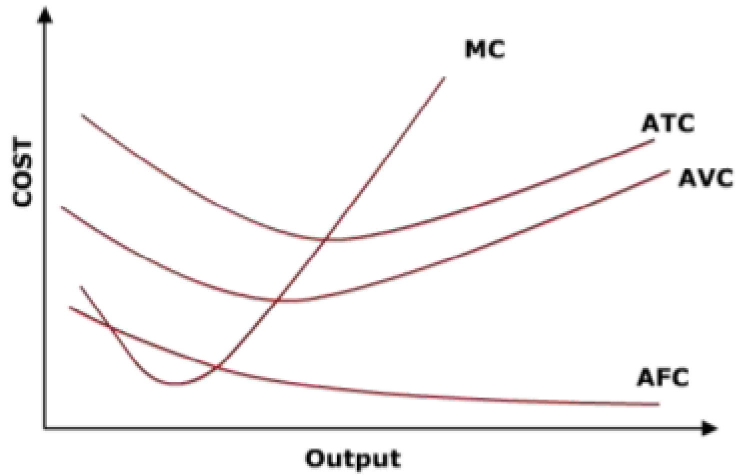
The relationship between AC and MC is as follows:

When MC curve is below the AC curve, the AC falls

When MC curve is above the AC curve, the AC rises

The MC curve intersects the AC curve at its minimum point.

The MC curve intersects AVC curve and AC curve at their minimum points.



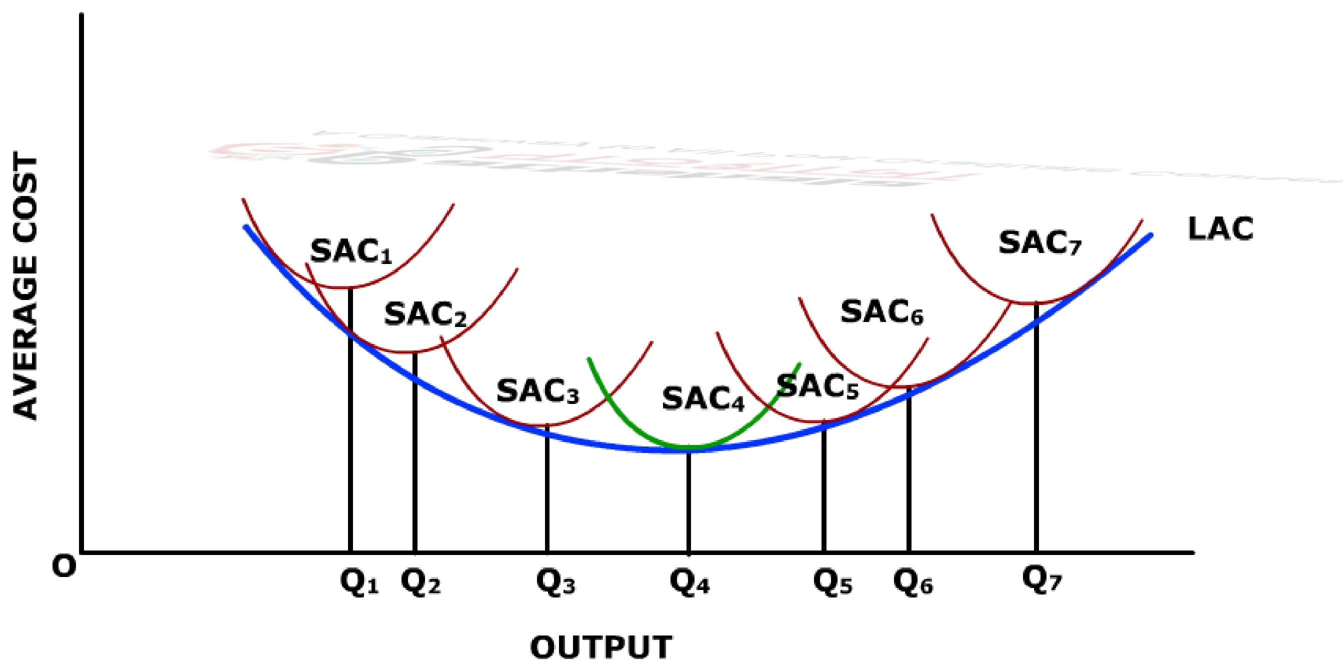
Long-Run Costs of Production (Analysis of Long Run AC curves):

In the long-run, all factors of production are variable. All costs are variable costs. The firm can alter the size and scale of the plant to meet the changed market conditions. Therefore, there is LAC curve and LMC curve. Long- run costs are known as PLANNING COSTS as the firm has to plan its future expansion of output. The long-run cost curves are subject to the law of returns to scale as the scale of production may undergo change.

LONG-RUN AVERAGE COST CURVE:

The Long –run Average cost curve or the LAC curve is the locus of points denoting the least cost of producing different levels of output in the long run. It shows the minimum average costs of producing the corresponding output in the long-run. **The LAC curve is thus a planning curve that guides the firm in deciding on the most optimal size of the plant for producing a given level of output.** An optimal sized plant is one which enables the production of the output at the minimum costs per unit of output. Given the technology, the firm is free to choose the plant size which entails the least cost.

The long-run average curve does not touch the short-run average cost curves on their minimum points. Graphically it can touch the minimum points of SACs only under constant returns to scale. In the phase of **increasing returns to scale and decreasing cost,** the LAC curve touches the SAC curves to the left of the minimum points of the SAC curves and in the phase of diminishing returns it touches the SAC curves to the right of their lowest point. The LAC is therefore, not the locus of lowest points of SAC curves. **The downward sloping portion of LAC comprises of only increasing returns and diminishing cost portions of SAC curves and vice versa.** The LAC curve is also known as the '**Envelope curve**' as it envelops the short run cost.



LONG-RUN AVERAGE COST CURVE

The Shape of the LAC: Economies and Diseconomies of Scale:

The shape of the long-run average cost depends on certain advantages and disadvantages associated with large scale production. These are known as economies and diseconomies of scale.

Economies of Scale:

Various factors may give rise to economies of scale, that is, to decreasing long-run average costs of production.

Greater Specialization of Resources:

With an expansion of a firm's scale of operation, its opportunities for specialization—whether performed by men or by machines—are greatly enhanced. It is because a large-scale firm can often divide the tasks and work to be done more readily than a small-scale firm.

More Efficient Utilization of Equipment:

In some industries, the technology of production is such that a large unit of costly equipment has to be used. The production of automobiles, steel and refined petroleum are obvious examples.

In such industries, companies must be able to afford whatever equipment is necessary and must be able to use it efficiently by spreading the cost per unit over a sufficiently large volume of output. A small-scale firm cannot ordinarily do these things.

Reduced Costs of Inputs:

A large-scale firm can often buy its inputs-such as its raw materials-at a cheaper price per unit and thus gets discounts on bulk purchases. Moreover, for certain types of equipment, the price per unit of capacity is often much less than larger sizes purchased.

For instance, the construction cost per square foot for a large factory is usually less than that for a small one. Again, the price per horsepower of various electric motors varies inversely with the amount of horsepower.

Diseconomies of Scale:

With continuous expansion of the scale of operation of a firm, a point may ultimately be reached when diseconomies of scale begin to exercise a more than offsetting effect on the firm's cost curve. As a result, the long-run average cost curve starts to rise.

This is attributable to the following two main reasons:

Decision-Making Role of Management:

As a firm becomes larger, heavier burdens are placed on the management so that eventually this resource input is overworked relative to others and 'diminishing returns' to management set in. In fact, management is an indivisible input which is not capable of continuous variation. With increase in the size of organisation there occurs delay in decision-making.

Competition for Resources:

Rising long-run average costs can occur as a growing firm increasingly bids labour or other resources away from other industries. In the real world, it is very difficult, if not virtually impossible, to determine just when diseconomies of scale are encountered and when they become strong enough to outweigh the economies of scale.